

Q1

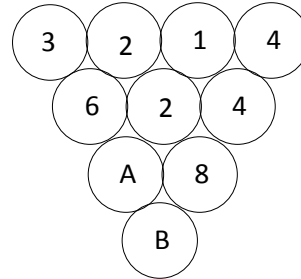
A palindrome year is a number that reads the same backward as forward.

2002 is an example.

Find the sum of the next two palindromes after 2002.

5

Q2



B = ?

5

Q3

Your Mom bakes a rectangular cake with length 25 cm width 12 cm and height 4 cm.

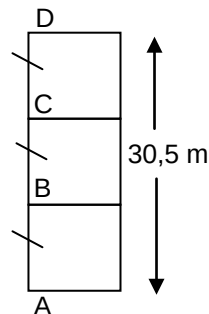
She then tops the cake with cream 1 cm thick.

What is the volume of the finished cake in cm^3 ?

5

Q4

An athlete sprints from A to B and back, then from A to C and back and finally from A to D and back.



How far did he sprint in TOTAL?

5

Q5

Find the average of the four prime numbers that are closest to 20.

5

Q6	In cooking 4 tablespoons = $\frac{1}{4}$ cup 1 cup = 250 ml. How many tablespoons are there in a LITRE?	Q7	In South Africa we measure speed in km per hour. In England they measure in miles per hour. 5 miles = 8 km. What is 120 km/h in miles/h?
	5		5
Q8	Jane's school is 50 km from her home. She must be at school by 08h00. If she travels at 50 km/h and arrives at 08h20, what time did she leave home?	Q9	I have an average of 78% for my first three tests in Mathematics. What mark must I get in my fourth test so that my overall average is 80%?
	5		5
	Q10	Today's date in the format dd/mm/yy is 30/01/14. Using ONLY the 6 digits in today's date, how many <u>OTHER</u> valid dates can be made that can happen this year?	
		5	

Q11 20÷14 = 1,428571428571.... has a recurring string of decimal digits. What digit is in position 2014 to the right of the decimal comma? 10	Q12 3 people can complete a task in 6 hours. How long would it take 4 people to complete the task if they work at the same rate as the original group of 3? 10
Q13 The height of a rectangle is <u>increased</u> by 20% and the base is <u>decreased</u> by 20%. By what % is the area of the original rectangle decreased? 10	Q14 1 @ 1 = 2 4 @ 6 = 52 5 @ 2 = 29 3 @ 8 = ? 10
Q15 5 2 1 7 3 1 6 A 5 B B = ? 10	

Q16 You can pick any 3 digits from the set 2 5 5 8 8 How many different 3-digit numbers can you make? 20	Q17 Two sides of a triangle measure 5 and 12 cm. The length of the third side side is a <u>NATURAL</u> number. How many possible triangles are there? 20
Q18 A jar contains R1 coins. 6 students try to guess how many coins are in the jar. The guesses are 42, 49, 52, 59, 62 and 65. One guess is 12 away from the correct answer; the other guesses are 1, 4, 6, 9 and 11 away. How many coins are in the jar ? 20	Q19 An island uses fruit as currency. 50 bananas = 20 apples 30 apples = 12 oranges 100 oranges = 1 peach How many bananas equals 1 peach? 20
Q20 If the arrangement of numbers shown below continues, what number would be directly below 144 ? 1 234 56789 10111213141516 [It may help to look at the last number in each row.] 20	

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JUNIOR (2014) [Grades 8 & 9]

Q1 4334	Q2 96	Q3 1500	Q4 122	Q5 18	 5 each
Q6 64	Q7 75	Q8 07h20	Q9 86	Q10 5	 5 each
Q11 5	Q12 4,5	Q13 4	Q14 73	Q15 3	 10 each
Q16 18	Q17 9	Q18 53	Q19 625	Q20 168	 20 each